

4/2/25 pre-party

I didn't realize just how far back the conceptual roots of artificial intelligence went. Even though it wasn't formally codified until the Dartmouth Conference in 1956, and even there it was simply introduced as a broad research topic that was simultaneously explored by multiple people with diverse interests.

4/2/25 reflection

Seeing these building blocks for search algorithms and such, and with the context from the readings, I wonder if there are similar primitive foundations for these concepts in some other facet of human history, from which these automated concepts are born.

Side note: difference between AI and Human is that human has autonomous goal. Not just a set of rules

Agent Taxonomy

Simple - reflex agents

Current action depends solely on current percept

Model-based agents

History of percepts used to build internal world model

Goal-based agents

Agent models a target state of the world it wants

Utility-based agents

to be in

Agent models the cost/benefit of a given state

Intelligent Agent: attempts to take "best" action

Rationality: Rational agent is one that does the right thing

Depends on: Performance measure that defines success criterion

Agent's prior knowledge of environment

Actions agent can perform

Agent's percept sequence to date

For each percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

PEAS Methodology

Before approaching any AI problem, first decide

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Before next time

Performance, Environment, Actuators, Sensors

Mind-body problem: Sometimes mind has separate objective than body
mind = software, body = hardware

LOL. ASIT 9/2

Pre-History of AI

Aristotle defined early logic

Descartes - I think therefore I am: I think $\rightarrow$ I Am

Also Dualism/rationalism

Leibniz built an early calculator, late 17th century

Babbage made a computer, Lovelace made an algorithm

Gödel found incompleteness decision trees

Good old-fashioned AI

Expert Systems: rule-based reasoning based on propositional/predicate calculus

History of AI

Dartmouth workshop in 1956

established AI as a research field

Early movements 52-69

Theorem Proving

Early Neural Networks (Perception)

Knowledge-based systems 69-79

Perceptions

Knowledge vs data

Human authored rules: Expert Systems

AI winter 1970's

coincided w/ knowledge based systems

Error Back-propagation for multi-layer networks 80's

Rediscovered Bryson/Ho

AI winter 90's until more layers/units deployed

00's and 10's: Big data, deep learning

20's: Deep learning, trustworthy AI, human centered, interpretable

Problem Solving w/ search

An agent can "look ahead"

Then, find sequence of actions to get to goal state

Eventually it gets there

Known as Problem-Solving Agent

This method is called search

Search

Simple Agent, Fully observable, Deterministic, known

informed search

Strat that uses problem-specific knowledge to find solution

Agent can estimate goal distance

Uninformed search

Strat w/o info for problem

no goal estimate

Search in Deterministic environment

Next state determined by current state and action

Search in Non-deterministic environment

guess.

Problem - Solving process

Goal Formulation

Problem Formulation

desc. of States/Actions

Search

Simulates sequence of possible actions

stops when sequence of actions to reach goal is found

otherwise stops when no solution is found

Execution

takes the actions it sought out

Search

States, initial state, goal states, actions

Often shown in weighted graphs, weights are actions

Actions

given state s , $Actions(s) \rightarrow$ returns a finite set of legal
~~actions~~ states in state s

Transition model

Cost function